

MODULE-1

Session 2

(TASKS)

1. What are the three main layers of software architecture? Write one line about each.

Ans:

- **Presentation Layer (UI Layer): -**

This layer provides the interface through which users interact with the software application.

- **Business Logic Layer (Application Layer): -**

This layer contains the core functionality and decision-making processes of the application.

- **Data Access Layer (Database Layer): -**

This layer is responsible for storing, retrieving, updating, and managing data in databases.

2. Define client and server in your own words with one real-life example.

Ans:

- **Client: -**

A client is a device, application, or software that sends requests to another system to access data or services. It acts as the user-facing part of a networked system.

- **Server: -**

A server is a computer or software that receives requests from clients and provides the requested information, services, or resources.

- **Real-Life Example: -**

When you use an ATM to check your bank balance, the ATM acts as the **client**, while the bank's central computer system acts as the **server** that processes the request and sends the account information.

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3. Write 2 differences each between desktop apps, web apps, and mobile apps.

Ans:

Desktop Apps	Web Apps	Mobile Apps
Usually offer higher performance because they run directly on the computer's operating system.	Depend on a web browser and internet connection for access.	Optimized for touch-screen devices and mobile usage.
Can work without an internet connection in many cases.	Updates are managed centrally and are instantly available to all users.	Can access mobile device features such as camera, GPS, and sensors.

4. Name any 3 databases and mention whether each is SQL or NoSQL.

Ans:

- **Oracle Database – SQL Database: -**

Oracle Database is a relational database system widely used in large enterprises for managing structured data.

- **Cassandra – NoSQL Database: -**

Cassandra is a distributed NoSQL database designed to handle large volumes of data across multiple servers.

- **SQLite – SQL Database: -**

SQLite is a lightweight relational database commonly used in mobile applications and embedded systems.